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SUBJECT / ASUNTO LOADS CORRECTION FACTOR IN GFEM MODELS WITH SKINS STIFFENED BY 1D ELEMENTS		From / De Gustavo Baños Dpt. / Dpto. Stress Methods and Opt. Approved / Aprobado Eduardo Oslé																																
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TEXT/TEXTO This memorandum is issued to confirm that the internal loads in upper and lower skins of lift boxes, obtained from GFEM models in which stringers are modelled by means of CBAR elements with correct offset values, need no further corrections in order to account for the real stiffness of the box section. <u>TABLE OF CONTENTS</u> <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 5%;">1</td> <td style="width: 90%;">INTRODUCTION.....</td> <td style="width: 5%; text-align: right;">2</td> </tr> <tr> <td>1.1</td> <td>OBJECT</td> <td style="text-align: right;">2</td> </tr> <tr> <td>1.2</td> <td>ABBREVIATIONS AND SYMBOLS</td> <td style="text-align: right;">2</td> </tr> <tr> <td>1.3</td> <td>REFERENCES</td> <td style="text-align: right;">2</td> </tr> <tr> <td>2</td> <td>DISCUSSION.....</td> <td style="text-align: right;">3</td> </tr> <tr> <td>2.1</td> <td>DESCRIPTION OF USE CASE</td> <td style="text-align: right;">3</td> </tr> <tr> <td>2.2</td> <td>LOAD CORRECTION FACTOR</td> <td style="text-align: right;">4</td> </tr> <tr> <td>2.3</td> <td>GFEM CONFIGURATIONS</td> <td style="text-align: right;">4</td> </tr> <tr> <td>2.4</td> <td>SUMMARY OF CHANGES OBSERVED IN GFEM</td> <td style="text-align: right;">9</td> </tr> <tr> <td>3</td> <td>CONCLUSSION.....</td> <td style="text-align: right;">9</td> </tr> </table>					1	INTRODUCTION.....	2	1.1	OBJECT	2	1.2	ABBREVIATIONS AND SYMBOLS	2	1.3	REFERENCES	2	2	DISCUSSION.....	3	2.1	DESCRIPTION OF USE CASE	3	2.2	LOAD CORRECTION FACTOR	4	2.3	GFEM CONFIGURATIONS	4	2.4	SUMMARY OF CHANGES OBSERVED IN GFEM	9	3	CONCLUSSION.....	9
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1 INTRODUCTION

1.1 OBJECT

A natural interpretation of Airbus DS applicable policies for creation of GFEM models of lift boxes is that the internal loads (in upper and lower skins) resulting of the FE analysis must be corrected by a dedicated factor, before using these loads in strength checks, even if stringers are modelled with (MSC Nastran) CBAR elements including a valid offset value (see Ref. 1 - §8.1 or the more recent Ref. 2 - §4.1).

Abovementioned correction factor intends to correct the error committed when positioning the 1D elements representing the stringers on the same grids used to mesh the skins surfaces, then overestimating the bending inertia of the box sections in the GFEM.

It results counter-intuitive that skins loads obtained with GFEM's where the stringers are modelled by means of CBAR elements, with offsets representative of the position of the centre of gravity of their sections, need any further correction for that reason, as the FE solver is expected to place the stringer inertia a its right place. This memorandum clarifies that this last rational is correct and no correction factors are needed in such cases (before the wordings of the formal policies are corrected in future revisions).

1.2 ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
FE	-	Finite Elements
GFEM	-	Global Finite Element Model
LEF	-	Load Enhancement Factor
N_x	N/mm	Load per unit length in longitudinal/axial (wing span) direction

1.3 REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	DPI-ADC-006	Ed. D	CRITERIOS GENERALES DE MODELIZACIÓN POR ELEMENTOS FINITOS PARA MSC.NASTRAN
Ref. 2	TEAS-NT-1810005	Is. A	GUIDELINES FOR GLOBAL FINITE ELEMENT MODELLING IN MSC.NASTRAN
Ref. 3	[Airbus DS]	10/21	Stress Methods Manual

Software used for the preparation of this document:

- MSC *Nastran* version 2016
- MSC *Patran* version 2018
- Airbus DS *SAM* version 1.0.0 (Classical Laminare Analysis module)

2 DISCUSSION

As noted above, it will be expected that the FE solver (MSC Nastran) interprets correctly the offset values of CBAR elements modelling stringers, and calculates with sufficient precision the inertia and stiffness of the box section in GFEM models. In order to confirm this expectation on the behaviour of this type of FE element, a representative use case has been defined, and then analysed with different type of stringer modellizations to collect and address the deviations.

2.1 DESCRIPTION OF USE CASE

A prismatic lift box made of a composite CFRP material, with 5 longitudinal tee stringers at lower and upper skin, has been used for this study. The box is supported as a cantilever beam (one end clamped, the other free), loaded at its free end with a bending moment causing tension in upper skin.

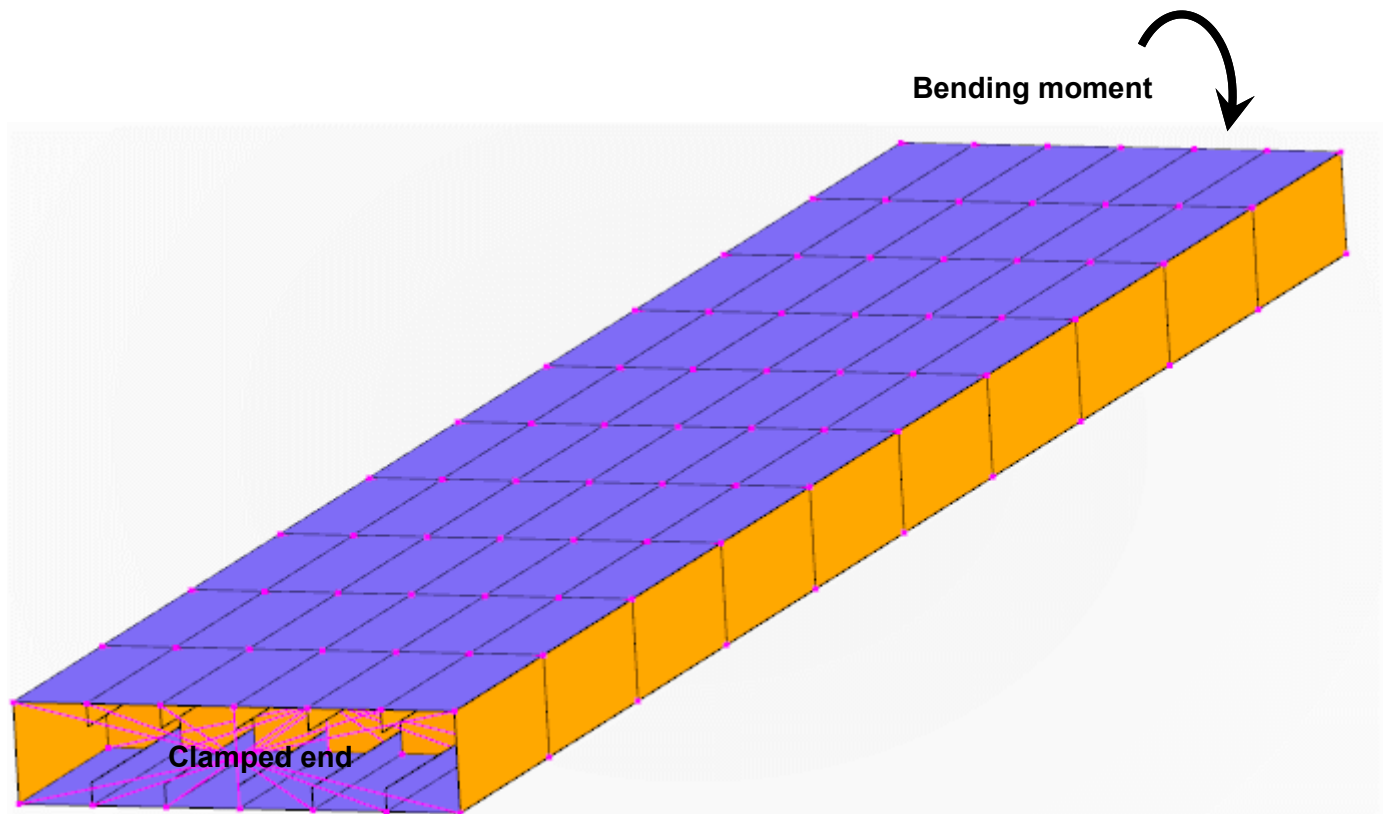


Figure 1. Sketch of the lift box used for this study

Box section dimensions

Box height: $H = 200 \text{ mm}$

Box width: $C = 900 \text{ mm}$

Stringer pitch $b = 150 \text{ mm}$

Tee stringer height: $h = 45 \text{ mm}$ (no foot, assumed integrated in skin)

Material and laminates

Material : CFRP pre-preg Hexcel IMA/M21E UD-Tape

t_{ply} (mm)	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12} -
0.184	154000	8500	4200	0.35

Table 1. Design values from the material datasheet in Airbus DS – SMM

Skin: Stacking (19 plies) (45/-45/90/90/90/45/-45/0/90/90)\$
 Thickness $t = 3.496 \text{ mm}$
 Elastic modulus $E_x = 36617 \text{ MPa}$

Stringer web: Stacking (25 plies) (45/-45/45/-45/0/90/0/0/0/90/0/90/0)\$
 Thickness $t = 4.600 \text{ mm}$
 Elastic modulus $E_x = 81548 \text{ MPa}$

Loads

Bending moment $2E7 \text{ Nmm}$

2.2 LOAD CORRECTION FACTOR

The method proposed in applicable stress policies for calculating that load correction factor in GFEM built with 1D elements modelling stringers would result for this use case in...

- Stringer offset $o = h / 2 = 45/2 \text{ mm} = 22.5 \text{ mm}$
- Stringer axial inertia $EA_{strg} = (E_x \cdot t \cdot h)_{st} = 81548 \cdot 4.600 \cdot 45 \text{ N} = 16.88E6 \text{ N}$
- Skin axial inertia $EA_{skin} = (E_x \cdot t \cdot b)_{sk} = 33617 \cdot 3.496 \cdot 150 \text{ N} = 17.63E6 \text{ N}$
- Offset of super-stinger elastic axis $o_{ss} = o \cdot EA_{strg} / (EA_{strg} + EA_{skin}) = 11.01 \text{ mm}$
- Correction factor $LEF = H / H' = H / (H - 2 \cdot o_{ss}) = 200 / (200 - 2 \cdot 11.01) = 1.12$

2.3 GFEM CONFIGURATIONS

Four different configurations of GFEM, representative of the prismatic box of this case of study, have been prepared to assess the differences in resulting internal loads. These models have the same general arrangement sketched in Figure 1, differenced only in the modelization of the longitudinal stringers in lower and upper skins:

- ✓ the 1st configuration uses CROD elements,
- ✓ the 2nd configuration uses CBAR elements without offset,

- ✓ the 3rd uses CBAR elements with an offset value equal to the distance from the stringer centres of gravity to the skin,
- ✓ and finally the reference configuration uses 2D elements normal to the skin

For simplicity, the loads monitored for this study are just the longitudinal/axial loads in a section of the lift box centred in the span.

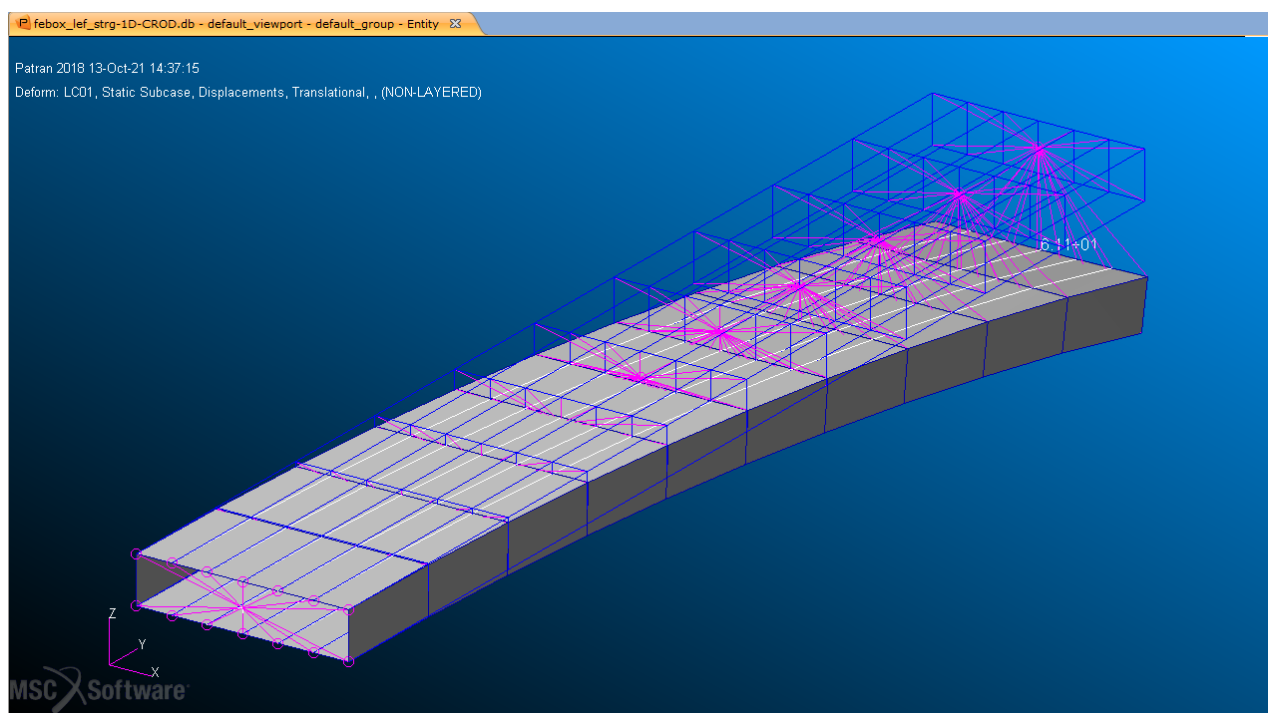
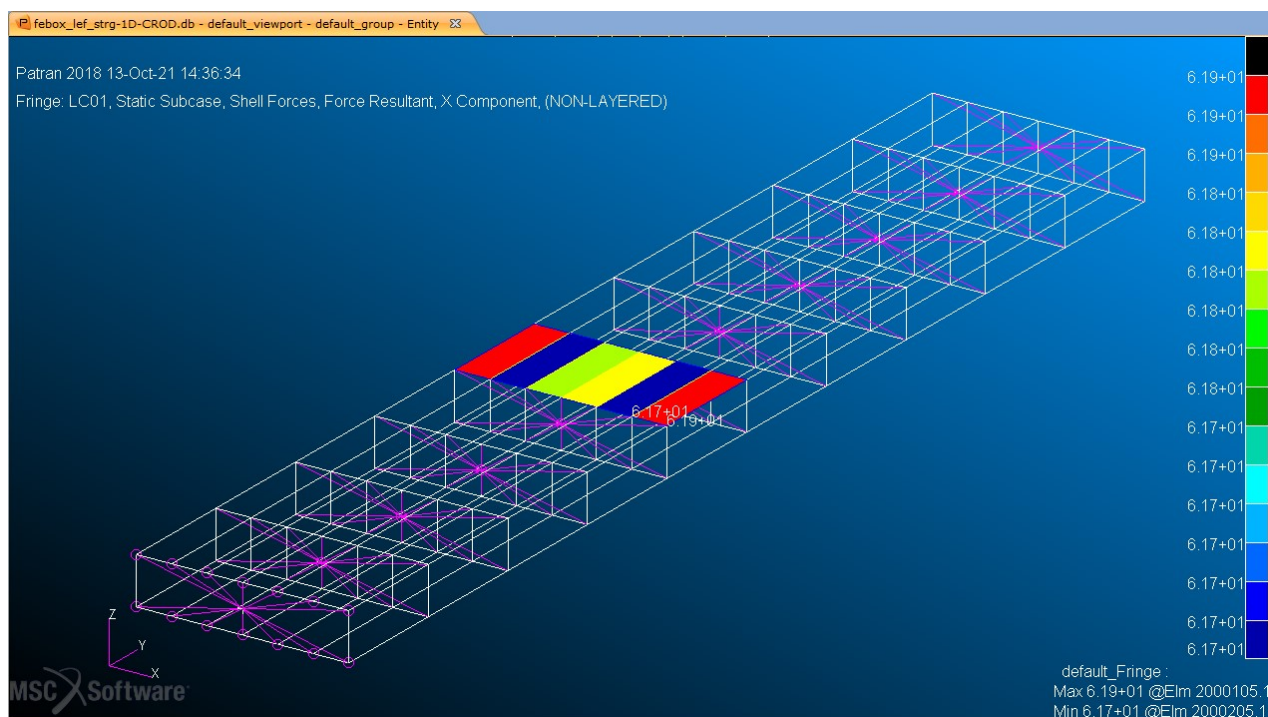


Figure 1. Stringers as CROD

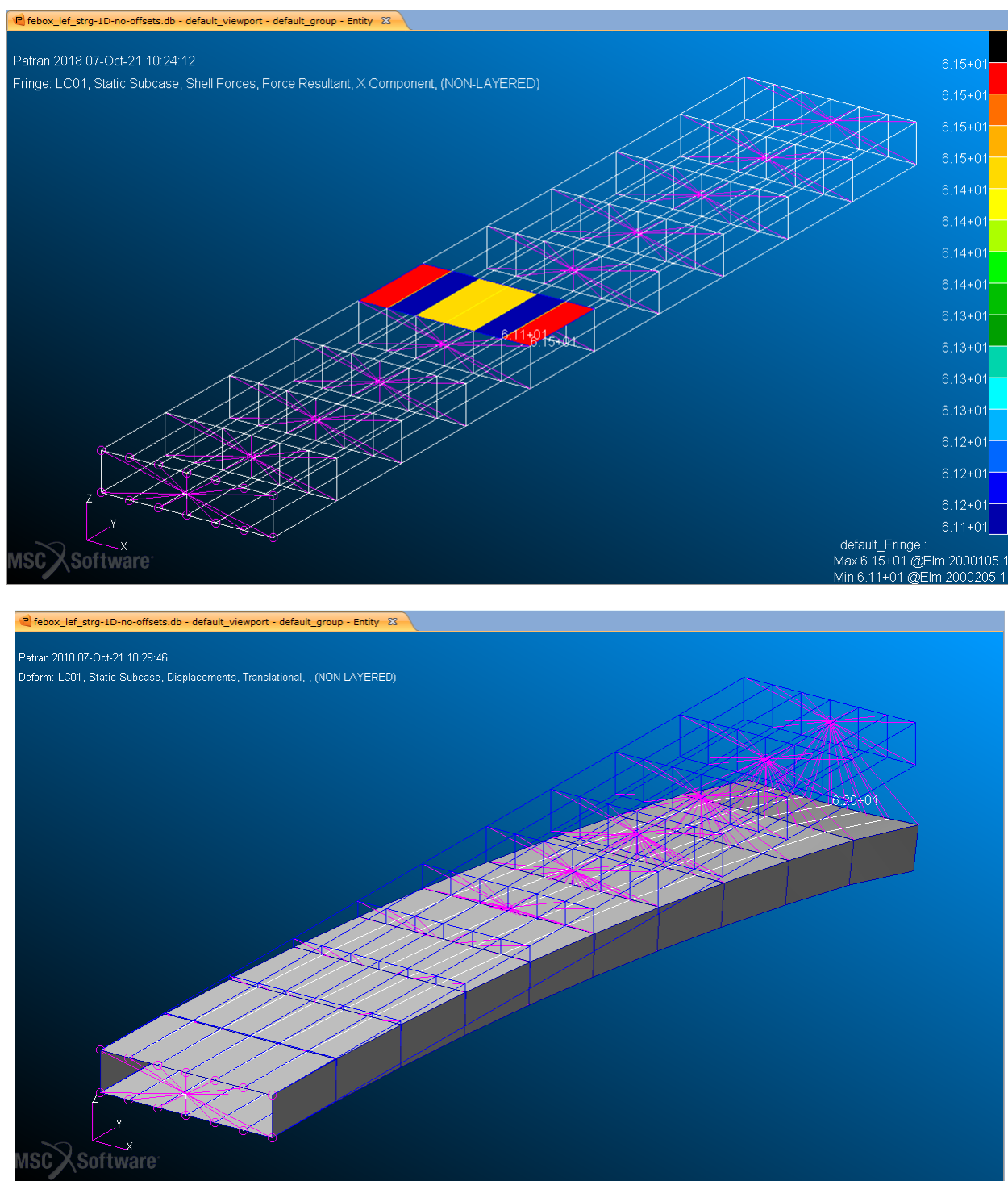


Figure 2. Stringers as CBAR without offsets

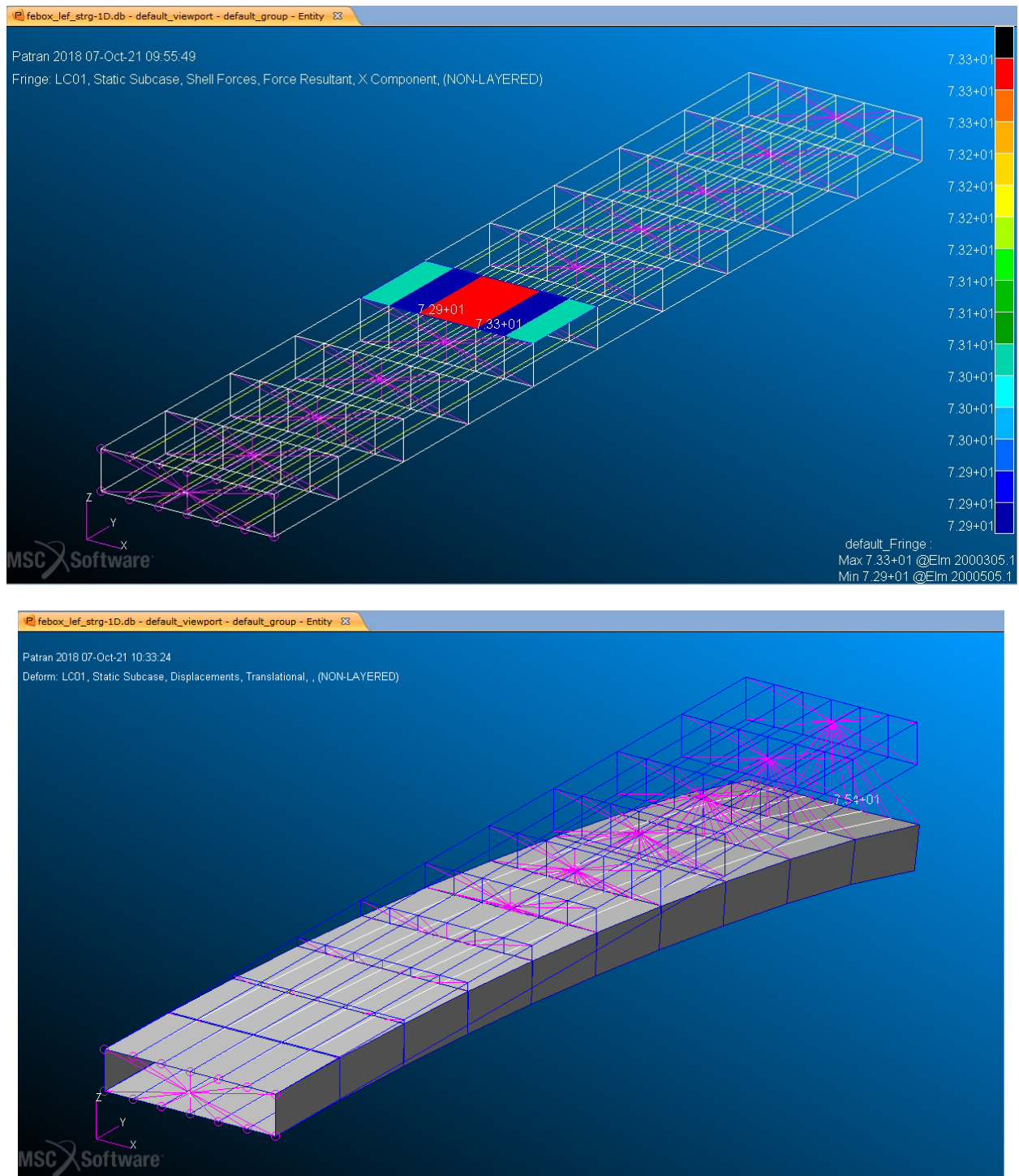


Figure 3. Stringers as CBAR with offsets

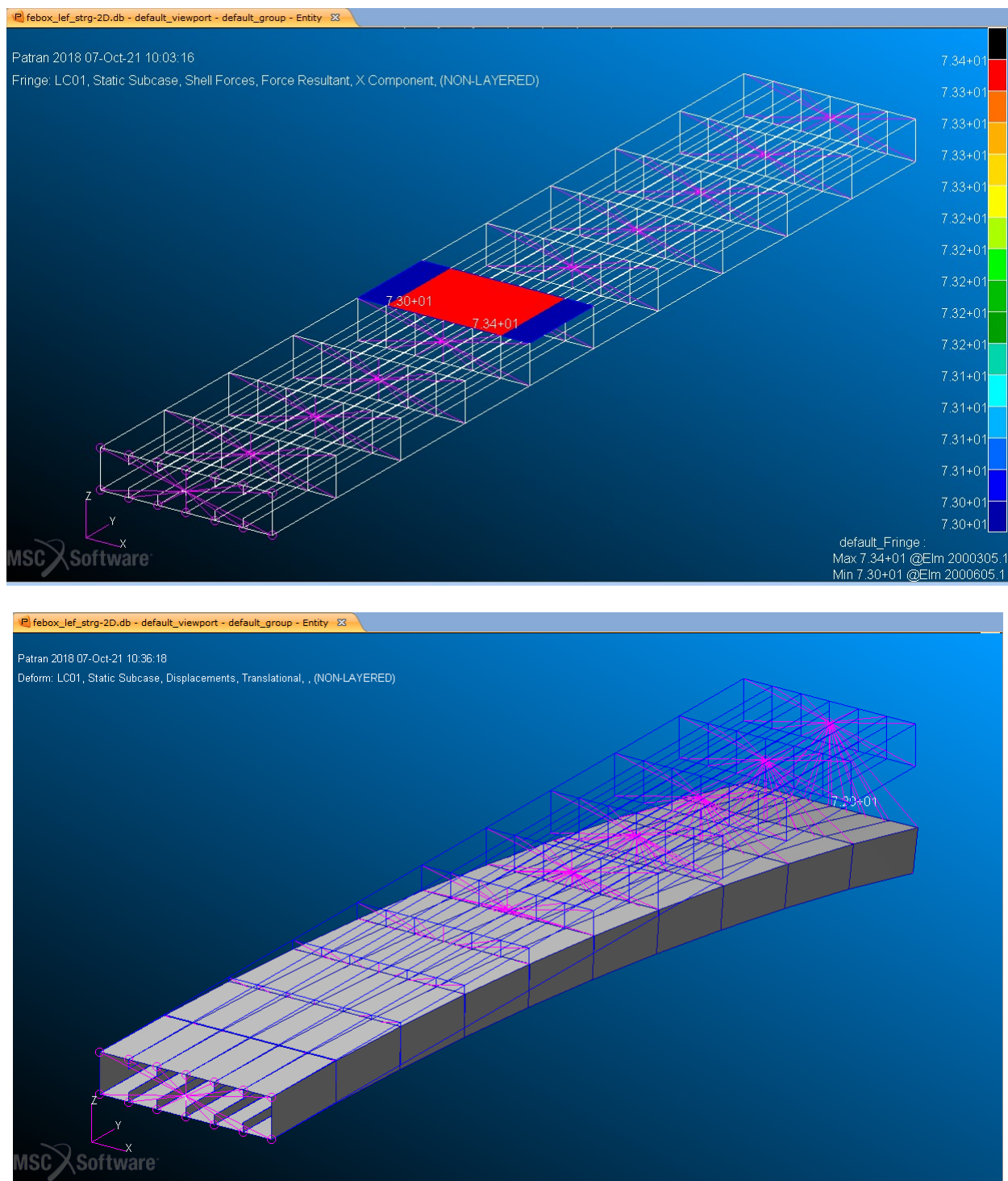


Figure 4. Stringers as CQUAD elements

2.4 SUMMARY OF CHANGES OBSERVED IN GFEM

The following table summarizes the GFEM results obtained at the monitoring stations and the correction factors which could be derived from those results.

<i>Stringer in GFEM</i>	<i>Nx (N/mm)</i>	<i>LEF</i>
CQUAD	7.3	1.00 (reference)
CBAR with offset	7.3	1.00
CBAR with no offset	6.1	1.20
CROD	6.2	1.18

Table 2. Observed load correction factor in GFEM's

3 CONCLUSION

From above discussion and analyses on the selected use case, the following conclusions can be extracted:

- ✓ No loads correction factor are needed if longitudinal stringers are modelled with CBAR elements with offset values representative of the stringer eccentricity
- ✓ If the stringers have sufficient eccentricity respect to the skin surface, the error committed using 1D elements with no offset can be difficult to evaluate (1.12 calculated vs 1.20 observed in this case), and this option should not be selected for the case. Instead of that, the CBAR modelization with offset is preferred, as actually stated in stress policies, Ref. 1 and Ref. 2, when calculated values of the correction factor are greater than 1.0.